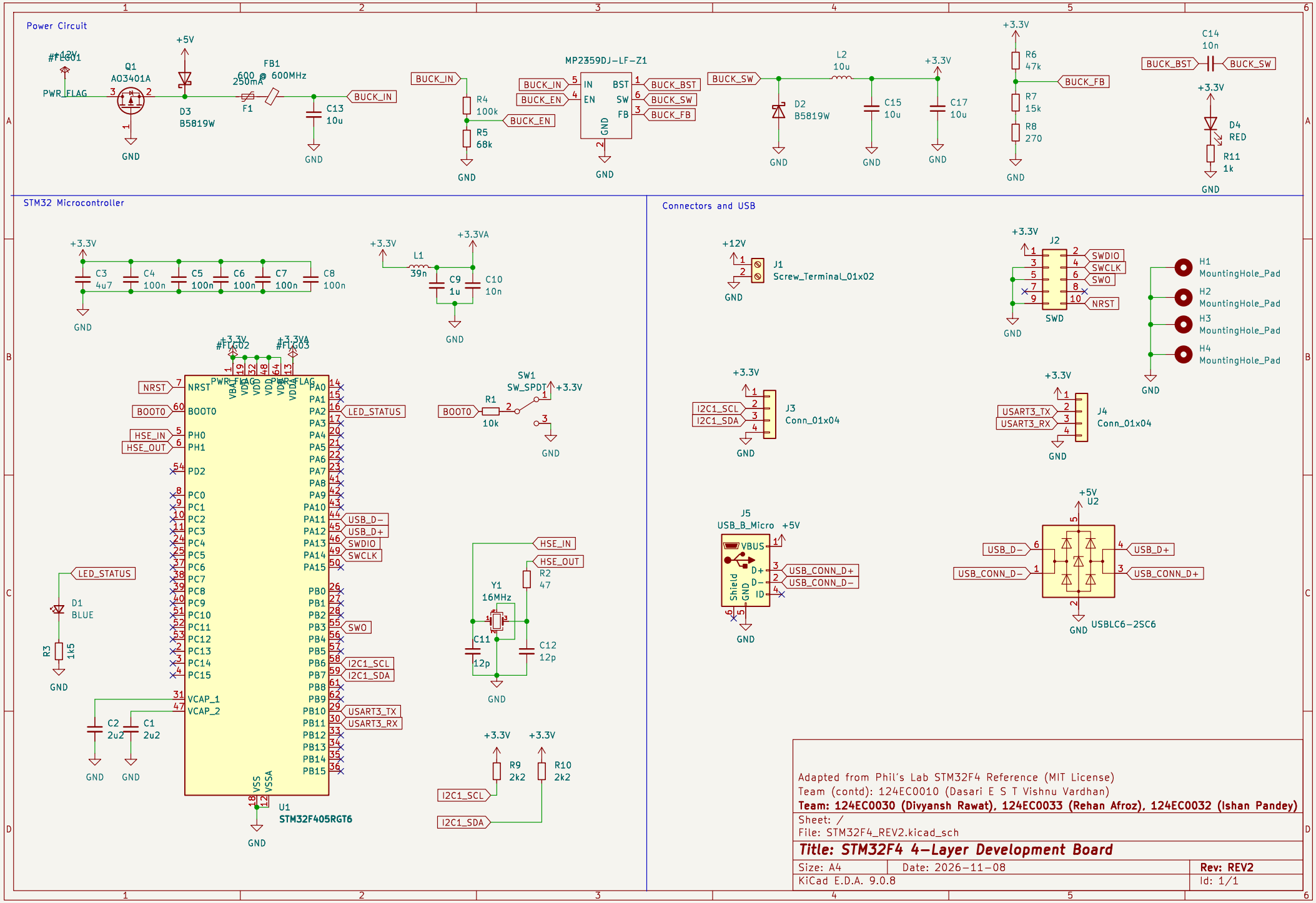

Power Circuit
STM32 Microcontroller
Connectors and USB
Q1
AO3401A
PWR_FLAG
D3
B5819W
FB1
600 @ 600MHz
250mA
F1
C13
10u
BUCK_IN
R4
100k
R5
68k
BUCK_EN
MP2359DJ-LF-Z1
BUCK_SW
L2
10u
D2
B5819W
C15
10u
C17
10u
R6
47k
R7
15k
R8
270
BUCK_FB
C14
10n
BUCK_BST
D4
RED
R11
1k
C3
4u7
C4
100n
C5
100n
C6
100n
C7
100n
C8
100n
L1
39n
C9
1u
C10
10n
U1
STM32F405RGT6
NRST
BOOT0
HSE_IN
HSE_OUT
LED_STATUS
USB_D-
USB_D+
SWDIO
SWCLK
SWO
I2C1_SCL
I2C1_SDA
USART3_TX
USART3_RX
SW1
SW_SPDT
R1
10k
D1
BLUE
R3
1k5
C2
2u2
C1
2u2
Y1
16MHz
C11
12p
C12
12p
R2
47
R9
2k2
R10
2k2
J1
Screw_Terminal_01x02
J2
SWD
J3
Conn_01x04
J4
Conn_01x04
J5
USB_B_Micro
USB_CONN_D+
USB_CONN_D-
U2
USBLC6-2SC6
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
Adapted from Phil's Lab STM32F4 Reference (MIT License)
Team (contd): 124EC0010 (Dasari E S T Vishnu Vardhan)
Team: 124EC0030 (Divyansh Rawat), 124EC0033 (Rehan Afroz), 124EC0032 (Ishan Pandey)
Sheet: /
File: STM32F4_REV2.kicad_sch
Title: STM32F4 4-Layer Development Board
Size: A4
Date: 2026-11-08
Rev: REV2
KiCad E.D.A. 9.0.8
Id: 1/1